

Problem 5 – Cracked section analysis

The cross section of a post-tensioned concrete beam spanning 16 m, simply supported, is shown in Figure 5.1. The beam is post-tensioned with a tendon, which has a parabolic profile, with end eccentricity zero and maximum eccentricity at midspan of 350 mm. The tendon comprises of 10 number 12.7 mm strands. The prestress force is 1325 kN.

Material properties are:

Concrete:

$$f'_c = 32 \text{ MPa}$$

$$f'_{cp} = 25 \text{ MPa (at 14 days when the tendon is stressed)}$$

$$\text{specific weight of concrete} = 25 \text{ kN} / \text{m}^3$$

Prestressing Strands:

$$\text{Cross-sectional area of a single strand} = 100 \text{ mm}^2$$

$$\text{Minimum breaking load of strands} = 184 \text{ kN}$$

$$E_p = 190,000 \text{ MPa}$$

Ordinary Reinforcement:

$$A_{sc} = 4N24 = 1800 \text{ mm}^2, \quad d_{sc} = 25 + \frac{20}{2} = 37 \text{ mm}$$

$$A_{st} = 8N20 = 3600 \text{ mm}^2, \quad d_{st} = 1000 - 25 + \frac{20}{2} = 963 \text{ mm}$$

If the beam is subjected to a factored service load of 50 kN/m (including self-weight):

1. Demonstrate that the beam is cracked under applied service loads
2. Perform a cracked section analysis to calculate the top fibre stress in concrete (assume that the neutral axis is in the web of the beam).

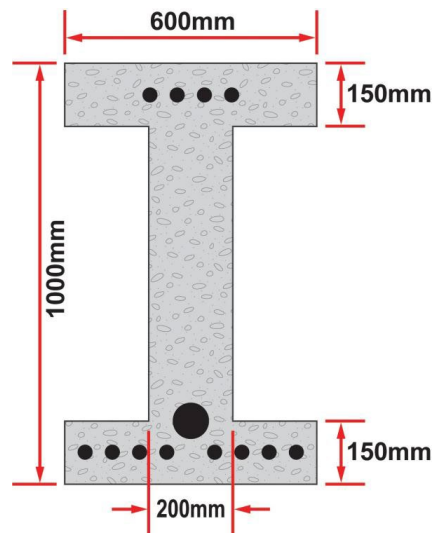


Figure 1.1 - Prestressed beam section